🚶 TOPIC 08.0: QUEUE FOUNDATION & FIFO PRINCIPLE

🔗 See Topic 07 → Stack Masterclass

🎯 **GOAL:** Master FIFO (First In, First Out) thinking, understand Double-Ended Queues (Deque), and master Monotonic Deque for Sliding Window Maximum in $O(N)$ time!

🧰 REUSABLE QUEUE & DEQUE UTILITIES

Queue & Deque Operation Reference:

```
// Queue (FIFO): offer(), poll(), peek()
Queue<Integer> q = new ArrayDeque<>();
q.offer(val); int front = q.poll();

// Deque (Double Ended): offerFirst(), offerLast(), pollFirst(),
pollLast()
Deque<Integer> deque = new ArrayDeque<>();
```

Circular Queue Modulo Math:

```
// Modulo Arithmetic for Circular Indexing
int nextIndex = (currIndex + 1) % capacity;
int prevIndex = (currIndex - 1 + capacity) % capacity;
```

💡 **AHA MOMENT #1: FIRST IN, FIRST OUT (FIFO) ORDER PRESERVATION**
A Queue is like a line at a supermarket checkout: elements enter at the rear (`enqueue / offer`) and leave from the front (`dequeue / poll`). Unlike Stack which reverses arrival order, Queue **preserves the exact arrival order**, making it the foundational engine for BFS graph traversals and streaming buffers!

1 CORE QUEUE OPERATIONS ($O(1)$ TIME)

- `offer(x)`: Adds element `x` to rear of queue ($O(1)$).
- `poll()`: Removes and returns front element ($O(1)$).
- `peek()`: Views front element without removing ($O(1)$).
- `isEmpty()`: Returns true if queue size is 0 ($O(1)$).

2 QUEUE VS STACK PARADIGM COMPARISON

Feature	Queue (FIFO)	Stack (LIFO)
Processing Order	First In, First Out	Last In, First Out
Primary Traversal	BFS (Level Order)	DFS (Depth First Search)
Entry/Exit Point	Enter Rear, Exit Front	Enter Top, Exit Top

🎨 VISUALIZING QUEUE ENQUEUE / DEQUEUE

1. OFFER(10), OFFER(20), OFFER(30): `FRONT [ 10 ⟶ 20 ⟶ 30 ] REAR` **2. POLL() ➔ Removes 10 (First Element):** `FRONT [ 20 ⟶ 30 ] REAR`

📌 **KEY TAKEAWAY:** Use Queue whenever you must process requests in order of arrival or explore graph nodes layer-by-layer!

🔄 TOPIC 08.1: CIRCULAR QUEUE & JAVA IMPLEMENTATIONS

💡 **AHA MOMENT #2: CIRCULAR ARRAY MODULO ARITHMETIC**
In a standard fixed array, dequeueing elements leaves empty spaces at index 0. Shifting remaining items costs $O(N)$! A **Circular Queue** re-uses empty spots using wrap-around modulo arithmetic:
`head = (head + 1) % capacity` and `tail = (tail + 1) % capacity` in $O(1)$ time!

1 CIRCULAR ARRAY QUEUE MECHANICS

```
class CircularQueue {
    private int[] data;
    private int head = 0, tail = 0, size = 0, capacity;
    public CircularQueue(int k) {
        capacity = k; data = new int[k];
    }
    public boolean enQueue(int value) {
        if (isFull()) return false;
        data[tail] = value;
        tail = (tail + 1) % capacity; // Modulo wrap
        size++; return true;
    }
    public boolean deQueue() {
        if (isEmpty()) return false;
        head = (head + 1) % capacity; // Modulo wrap
        size--; return true;
    }
    public boolean isEmpty() { return size == 0; }
    public boolean isFull() { return size == capacity; }
}
```

2 ARRAYDEQUE VS LINKEDLIST QUEUE

Feature	`ArrayDeque`	`LinkedList`
Memory Overhead	Contiguous Array (Low)	Heap Node Objects (High)
Cache Locality	High (CPU Cache friendly)	Low (Pointer chasing)
Null Values	Disallows `null`	Allows `null`
FAANG Verdict	✅ **PREFERRED CHOICE**	⚠️ Avoid unless null needed

⚠️ **JAVA API TIP:** Prefer `offer()`, `poll()`, `peek()` over `add()`, `remove()`, `element()` because offer/poll return `false` / `null` instead of throwing exceptions!

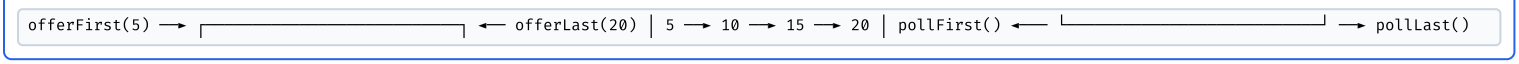
↔️ TOPIC 08.2: DEQUE FOUNDATION (DOUBLE-ENDED QUEUE) PAGE 3 OF 10

💡 **AHA MOMENT #3: DUAL-ENDED POWER (FRONT AND BACK ACCESS)**
A **Deque (Double-Ended Queue)** allows inserting and deleting from BOTH the front AND the back in $O(1)$ time! This versatility allows a Deque to act as a FIFO Queue, a LIFO Stack, or a **Monotonic Deque** for Sliding Window Maximum!

📒 DEQUE METHOD MAPPING CHEAT SHEET

Operation	Front Method	Back Method	Equivalent Stack/Queue Usage
Insert	`addFirst(e)` / `offerFirst(e)`	`addLast(e)` / `offerLast(e)`	Queue Enqueue: `offerLast()`, Stack Push: `push() / addFirst()`
Remove	`removeFirst()` / `pollFirst()`	`removeLast()` / `pollLast()`	Queue Dequeue: `pollFirst()`, Stack Pop: `pop() / removeFirst()`
Examine	`getFirst()` / `peekFirst()`	`getLast()` / `peekLast()`	Queue Front: `peekFirst()`, Stack Top: `peek() / peekFirst()`

🎨 VISUALIZING DOUBLE-ENDED QUEUE (DEQUE)



📌 **KEY TAKEAWAY:** Always declare `Deque<Integer> deque = new ArrayDeque<>();` for maximum efficiency!

📈 TOPIC 08.3: MONOTONIC DEQUE MASTERCLASS (SLIDING WINDOW MAX)

💡 **AHA MOMENT #4: MONOTONIC DECREASING DEQUE ($O(N)$ SLIDING WINDOW MAX)**
To find the Maximum in a Sliding Window of size K in $O(N)$ total time, maintain a **Monotonic Decreasing Deque of indices**:
1. **Expire Old Elements**: If `deque.peekFirst() ≤ i - K`, pop from FRONT!
2. **Maintain Decreasing Order**: While `!deque.isEmpty() && arr[i] ≥ arr[deque.peekLast()]`, pop from BACK!
3. Maximum for current window is ALWAYS at `deque.peekFirst()`!

📊 **STEP-BY-STEP DEQUE STATE TRACE: WINDOW = [1, 3, -1, -3, 5], K = 3**

Read i	Value nums[i]	Monotonic Deque Action	Deque State (Indices / Values)	Window Max (peekFirst)
0	1	Push index 0 ➔ Deque: `[0]`	`[1]`	N/A ($i < K-1$)
1	3	3 > 1 ➔ Pop 0 from BACK. Push 1 ➔ Deque: `[1]`	`[3]`	N/A ($i < K-1$)
2	-1	-1 < 3 ➔ Push 2 ➔ Deque: `[1, 2]`	`[3, -1]`	**3** (at idx 1)
3	-3	-3 < -1 ➔ Push 3 ➔ Deque: `[1, 2, 3]`	`[3, -1, -3]`	**3** (at idx 1)
4	5	Expire idx 1 (1 <= 4-3). 5 > -3, 5 > -1, 5 > 3 ➔ Pop all from BACK! Push 4	`[5]`	**5** (at idx 4)

📌 **KEY TAKEAWAY:** Monotonic Deque achieves $O(N)$ runtime because each element is pushed once and popped at most once!

📜 TOPIC 08.4, 08.5 & 1-PAGE PRINTABLE CHEAT SHEET

🌳 QUEUE & DEQUE INTERVIEW DECISION TREE (MERMAID)

📜 PRINTABLE QUEUE & DEQUE CHEAT SHEET

Pattern Type	Target Problem	Data Structure	Key Code Mechanism
Standard FIFO Queue	Recent Calls (LC 933)	`Queue<Integer> q = new ArrayDeque<>()`	`q.offer(val)`, `q.poll()`
Circular Queue	Design Circular Queue (LC 622)	Array + Modulo Pointers	`head = (head + 1) % K`
Sliding Window Max	Sliding Window Max (LC 239)	Monotonic Decreasing Deque	Pop smaller from back, pop expired from front
BFS Layer Processing	Level Order Traversal	Queue	`int size = q.size(); for(int i=0; i<size; i++)`
Two-Queue Voting	Dota2 Senate (LC 649)	2 Queues (Radiant, Dire)	Compare front indices, re-enqueue winner with `+N` offset

🎓 **FAANG COACH TIP:** Print this page and review it 10 minutes before your technical interview!

🧩 EASY NEETCODE: QUEUE USING STACKS, RECENT CALLS & MOVING AVG

1. LEETCODE 232 — IMPLEMENT QUEUE USING STACKS

Easy ★★★★★ **Two Stacks** Amazon Microsoft

Problem:

Implement FIFO queue using two stacks (`input` & `output`). Amortized $O(1)$ time!

```
class MyQueue {
    private Deque<Integer> in = new ArrayDeque<>();
    private Deque<Integer> out = new ArrayDeque<>();
    public void push(int x) { in.push(x); }
    public int pop() { peek(); return out.pop(); }
    public int peek() {
        if (out.isEmpty()) {
            while (!in.isEmpty()) out.push(in.pop()); // Reverse order
        }
        return out.peek();
    }
    public boolean empty() { return in.isEmpty() && out.isEmpty(); }
}
```

2. LEETCODE 933 — NUMBER OF RECENT CALLS

Easy ★★★★☆ **Queue Range Expiry** Google

Problem:

Count number of recent requests within time range `[t - 3000, t]`.

```
class RecentCounter {
    private Queue<Integer> q = new ArrayDeque<>();
    public int ping(int t) {
        q.offer(t);
        while (!q.isEmpty() && q.peek() < t - 3000) {
            q.poll(); // Expire timestamps older than t - 3000
        }
        return q.size();
    }
}
```

💡 **AHA MOMENT (LC 232):** Two LIFO reversals cancel out to form a perfect FIFO Queue!

3. LEETCODE 239 — SLIDING WINDOW MAXIMUM

Hard ★★★★★ Monotonic Decreasing Deque Amazon Meta Google

📌 FLOWCHART: Expire old idx from front ➔ Pop smaller from back ➔ Record peekFirst() when i ≥ K-1!

Problem:

Return maximum element in sliding window of size K sliding from left to right in $O(N)$ total time.

```
public int[] maxSlidingWindow(int[] nums, int k) {
    int n = nums.length, idx = 0;
    int[] res = new int[n - k + 1];
    Deque<Integer> deque = new ArrayDeque<>(); // Stores indices
    for (int i = 0; i < n; i++) {
        if (!deque.isEmpty() && deque.peekFirst() <= i - k) {
            deque.pollFirst(); // Expire index outside window
        }
        while (!deque.isEmpty() && nums[i] >= nums[deque.peekLast()]) {
            deque.pollLast(); // Maintain monotonic decreasing order
        }
        deque.offerLast(i);
        if (i >= k - 1) res[idx++] = nums[deque.peekFirst()];
    }
    return res;
}
```

4. LEETCODE 622 — DESIGN CIRCULAR QUEUE

Medium ★★★★☆ Array Modulo Pointer Amazon Meta

Problem:

Design a Circular Queue using a fixed-size array with $O(1)$ operations.

```
class MyCircularQueue {
    private int[] data;
    private int head = 0, tail = 0, size = 0, capacity;
    public MyCircularQueue(int k) { capacity = k; data = new int[k]; }
    public boolean enQueue(int value) {
        if (isFull()) return false;
        data[tail] = value; tail = (tail + 1) % capacity; size++; return
true;
    }
    public boolean deQueue() {
        if (isEmpty()) return false;
        head = (head + 1) % capacity; size--; return true;
    }
    public int Front() { return isEmpty() ? -1 : data[head]; }
    public int Rear() { return isEmpty() ? -1 : data[(tail - 1 +
capacity) % capacity]; }
    public boolean isEmpty() { return size == 0; }
    public boolean isFull() { return size == capacity; }
}
```

💡 **AHA MOMENT (LC 239):** Monotonic Deque cuts time complexity from $O(N \cdot K)$ brute-force to $O(N)$ optimal!

5. LEETCODE 649 — DOTA2 SENATE

Medium ★★★★☆ Two Queues Simulation Uber

Problem:

Predict winning Senate party (`Radiant` vs `Dire`) in voting round simulation.

```
public String predictPartyVictory(String senate) {
    int n = senate.length();
    Queue<Integer> rad = new ArrayDeque<>(), dir = new ArrayDeque<>();
    for (int i = 0; i < n; i++) {
        if (senate.charAt(i) == 'R') rad.offer(i);
        else dir.offer(i);
    }
    while (!rad.isEmpty() && !dir.isEmpty()) {
        int rIdx = rad.poll(), dIdx = dir.poll();
        if (rIdx < dIdx) rad.offer(rIdx + n); // Radiant bans Dire!
        else dir.offer(dIdx + n);             // Dire bans Radiant!
    }
    return rad.isEmpty() ? "Dire" : "Radiant";
}
```

6. LEETCODE 950 — REVEAL CARDS IN INCREASING ORDER

Medium ★★★★☆ Deque Index Simulation Google

Problem:

Order deck of cards so that revealing top card and moving next to bottom reveals in sorted order.

```
public int[] deckRevealedIncreasing(int[] deck) {
    int n = deck.length; Arrays.sort(deck);
    Deque<Integer> indexDeque = new ArrayDeque<>();
    for (int i = 0; i < n; i++) indexDeque.offer(i);
    int[] res = new int[n];
    for (int card : deck) {
        res[indexDeque.pollFirst()] = card; // Place smallest card
        if (!indexDeque.isEmpty()) {
            indexDeque.offerLast(indexDeque.pollFirst()); // Move next
to back
        }
    }
    return res;
}
```

💡 **AHA MOMENT (LC 649):** Re-enqueueing the winning senator with an index offset `idx + N` perfectly simulates round-robin voting!

7. LEETCODE 862 — SHORTEST SUBARRAY WITH SUM AT LEAST K

Hard ★★★★★ Prefix Sum + Monotonic Deque Amazon Google

Problem:
Length of shortest non-empty subarray with sum $\ge K$ (handles negative numbers!).

```
public int shortestSubarray(int[] nums, int k) {
    int n = nums.length, minLen = n + 1;
    long[] P = new long[n + 1];
    for (int i = 0; i < n; i++) P[i + 1] = P[i] + nums[i];
    Deque<Integer> deque = new ArrayDeque<>();
    for (int i = 0; i ⩽ n; i++) {
        while (!deque.isEmpty() && P[i] - P[deque.peekFirst()] ⩾ k) {
            minLen = Math.min(minLen, i - deque.pollFirst());
        }
        while (!deque.isEmpty() && P[i] ⩽ P[deque.peekLast()]) {
            deque.pollLast(); // Maintain monotonic increasing prefix
sums
        }
        deque.offerLast(i);
    }
    return minLen ⩽ n ? minLen : -1;
}
```

8. LEETCODE 1425 — CONSTRAINED SUBSEQUENCE SUM

Hard ★★★★★ DP + Monotonic Deque Google

Problem:
Max sum of subsequence such that index difference between consecutive elements is $\le K$.

```
public int constrainedSubsetSum(int[] nums, int k) {
    int n = nums.length, maxSum = nums[0];
    int[] dp = new int[n];
    Deque<Integer> deque = new ArrayDeque<>();
    for (int i = 0; i < n; i++) {
        if (!deque.isEmpty() && deque.peekFirst() < i - k)
deque.pollFirst();
        dp[i] = nums[i] + (!deque.isEmpty() ? Math.max(0,
dp[deque.peekFirst()]) : 0);
        maxSum = Math.max(maxSum, dp[i]);
        while (!deque.isEmpty() && dp[i] ⩾ dp[deque.peekLast()])
deque.pollLast();
        deque.offerLast(i);
    }
    return maxSum;
}
```

🎓 **FAANG COACH TIP:** Combining Dynamic Programming with a Monotonic Deque reduces $O(N \cdot K)$ DP transitions to $O(N)$ linear time!

💎 IF YOU REMEMBER ONLY ONE PAGE... MASTER REVISION SHEET

⭐ THE 4 GOLDEN LAWS OF QUEUES & DEQUES

1. **Use ArrayDeque for Queue & Deque:** Always declare `Deque<Integer> q = new ArrayDeque<>();` (Faster than LinkedList!).
2. **Monotonic Decreasing Deque for Sliding Max:** Pop smaller elements from BACK, pop expired indices from FRONT.
3. **Circular Queue Modulo Math:** Wrap indices via `(idx + 1) % capacity` to avoid $O(N)$ element shifting.
4. **BFS Queue Snapshot:** For layer-by-layer tree/graph processing, freeze size via `int size = q.size();` before inner loop!

⛔ WHEN QUEUE FAILS (DO NOT USE!)

- **Random Index Access:** Queue ONLY allows viewing front element ($O(1)$) ➔ Use **Array / ArrayList**!
- **LIFO Reversal:** Queue preserves order ➔ Use **Stack** for LIFO reversal!

📊 NEETCODE & BLIND 75 CONFIDENCE TRACKER

🟢 Easy LC 232 Queue using Stacks	🟢 Easy LC 933 Recent Calls	🟢 Easy Moving Average Stream
🟠 Med LC 622 Circular Queue	🟠 Med LC 649 Dota2 Senate	🟠 Med LC 950 Reveal Cards Order
🔴 Hard LC 239 Sliding Window Max	🔴 Hard LC 862 Shortest Subarray Sum	🔴 Hard LC 1425 Constrained Subset Sum

👉 **Next Master Topic: Topic 09 → Priority Queue & Heap Masterclass**

🎉 **TOPIC 08 QUEUE & DEQUE MASTERCLASS COMPLETE!** Turn the page to **Topic 09: Priority Queue & Heap Masterclass!**